WEEK 07 NETWORK ADMINISTRATION

Enterprise Network Security, Monitoring & Verification

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Tools Used: GNS3, Cisco IOS, PuTTY

Assignment: Week 07

Topic: Enterprise Network Security, Monitoring & Verification

Introduction

Week 07 focused on enterprise network security, monitoring, secure management, VPN implementation, routing security, and final network verification. The practical work was performed using GNS3 and Cisco IOS routers.

The purpose of this assignment was to secure the enterprise network against unauthorized access, protect routing information, establish secure communication through an IPsec VPN, configure network monitoring services, apply basic router hardening, and verify connectivity and security.

The main technologies covered in this assignment include IPsec VPN, IKE, SSH version 2, AAA, OSPF authentication, NTP, Syslog, SNMPv3, ACLs, basic hardening, and Zone-Based Firewall.

The report documents the configurations that were successfully completed and also clearly identifies the features that could not be implemented because of limitations in the available IOS image or topology.

Objectives

The main objectives of Week 07 were:

- Configure and verify an IPsec VPN.
- Configure IKE Phase 1.
- Configure IPsec Phase 2.
- Configure pre-shared key authentication.
- Configure a transform set.
- Apply a crypto map.
- Configure SSH version 2.
- Disable Telnet.
- Configure AAA authentication.
- Configure local user authentication.
- Enable password encryption.
- Configure a MOTD banner.
- Secure VTY lines.
- Configure OSPF authentication.
- Configure passive interfaces.

- Apply prefix-list filtering.
- Configure NTP.
- Configure Syslog/logging.
- Configure SNMPv3.
- Apply basic router hardening.
- Perform ping and traceroute testing.
- Verify the VPN.
- Verify ACL/security configuration.
- Configure Zone-Based Firewall where supported.

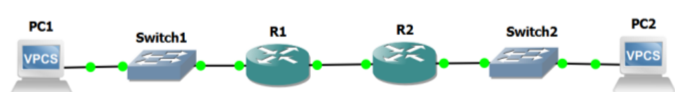
Network Topology

The Week 07 network topology used for the practical implementation was:

PC1 — Switch1 — R1 ═════ WAN ═════ R2 — Switch2 — PC2

R1 was used as the main edge/security router.

Topology



IP Addressing and Interface Verification

The interfaces on R1 were checked before applying the remaining security configurations.

R1 had two active interfaces:

- FastEthernet0/0 — 192.168.10.1 — LAN
- FastEthernet1/0 — 10.0.12.1 — WAN

Both interfaces were required for the topology and were in an operational state.

Verification Command

show ip interface brief

The verification confirmed that both interfaces were up and operational.

R1

IPsec VPN Security

IPsec VPN was configured to provide secure communication between the WAN peers.

The VPN implementation included:

- IKE Phase 1
- Pre-shared key
- IPsec Phase 2
- Transform set
- Crypto map

The configuration was completed before the final verification stage.

5.1 IKE Phase 1

IKE Phase 1 was configured to establish a secure security association between the VPN peers.

The following command was used to verify the IKE security association:

show crypto isakmp sa

The output showed:

- Source: 10.0.12.1

- Destination: 10.0.12.2
- State: QM_IDLE
- Status: ACTIVE

The **QM_IDLE** state indicates that the IKE security association reached its stable operational state.

IKE phase 1

```
R2#show crypto isakmp policy

Global IKE policy
Protection suite of priority 10
        encryption algorithm:   AES - Advanced Encryption Standard (128 bit keys).
        hash algorithm:         Secure Hash Standard
        authentication method:  Pre-Shared Key
        Diffie-Hellman group:   #5 (1536 bit)
        lifetime:               86400 seconds, no volume limit
R2#
```

IPsec Phase 2

IPsec Phase 2 was configured using a transform set and crypto map.

The transform set provides the security parameters for protected traffic, while the crypto map associates the VPN policy with the required interface.

IPsec Phase 2

```
R2#show crypto ipsec sa

interface: FastEthernet1/0
    Crypto map tag: VPN-MAP, local addr 10.0.12.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.20.0/255.255.255.0/0/0)
   remote ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
   current_peer 10.0.12.1 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 9, #pkts encrypt: 9, #pkts digest: 9
    #pkts decaps: 9, #pkts decrypt: 9, #pkts verify: 9
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 10.0.12.2, remote crypto endpt.: 10.0.12.1
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet1/0
     current outbound spi: 0xE119C093(3776561299)
     PFS (Y/N): N, DH group: none

     inbound esp sas:
      spi: 0x63BA45B4(1673151924)
 --More--
```

Secure Management and AAA

Secure router management was implemented using SSH version 2.

The following security features were completed:

- SSH version 2
- Telnet disabled
- AAA configuration
- Local username
- Password encryption
- MOTD banner
- Secure VTY lines

SSH provides encrypted remote management instead of insecure Telnet access.

AAA was configured to provide authentication and centralized control over administrative access.

Password encryption was also enabled to prevent passwords from appearing in plain text in the running configuration.

Secure SSH management, AAA, and VTY configuration.

```
R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded):
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAAAgQDMiKKc4P8R2h80JPqFfGSOp9EhlN/NOpe2ERBEBuKa
RhTmHjcfckMHvh0M5ih1/DQg/xPJomRQ6SO/llpVRhneOgoqnmx1BQDTd3vobNCDejJKxMQc96JR11VF
Gjtpg0YHXlb02XsM6cDP7tKYOCnVwR04Q4OzpeqP3m+yX0FWYw==
R1#
```

```
R1#show running-config | section aaa
aaa new-model
aaa authentication login default local
aaa session-id common
R1#
```

```
R1#show running-config | section line vty
line vty 0 4
 transport input ssh
R1#
```

OSPF Security

OSPF security was configured to protect routing information and reduce unauthorized routing participation.

The completed OSPF security features included:

- OSPF authentication
- Passive interface
- Prefix-list filtering

OSPF authentication helps protect routing adjacencies from unauthorized participation.

Passive interfaces prevent unnecessary OSPF hello packets from being sent toward interfaces where an OSPF neighbor is not expected.

Prefix-list filtering helps control which routes are permitted or advertised.

Passive Interface and prefix list

```
R2#show running-config | section router ospf
router ospf 1
 router-id 2.2.2.2
 passive-interface FastEthernet0/0
 network 10.0.12.0 0.0.0.255 area 0
 network 192.168.20.0 0.0.0.255 area 0
R2#
```

```
ip prefix-list WEEK7-FILTER: 2 entries
   seq 5 deny 10.99.99.0/24
   seq 10 permit 0.0.0.0/0 le 32
R2#
```

NTP Configuration

Network Time Protocol was configured to provide consistent time synchronization in the laboratory network.

R1 was configured as an NTP master, while R2 was configured to use R1 as its NTP server.

R1

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ntp master 3
R1(config)#end
R1#
Aug 14 21:42:17.310: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

```
R2#enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ntp server 10.0.12.1
R2(config)#end
R2#
*Aug 14 21:43:18.972: %SYS-5-CONFIG_I: Configured from console by console
R2#
```

This configuration allows R2 to obtain its time reference from R1.

Accurate time is important for network troubleshooting, security events, and log analysis.

Syslog / Router Logging

Syslog and router logging were addressed using the router's built-in logging capabilities.

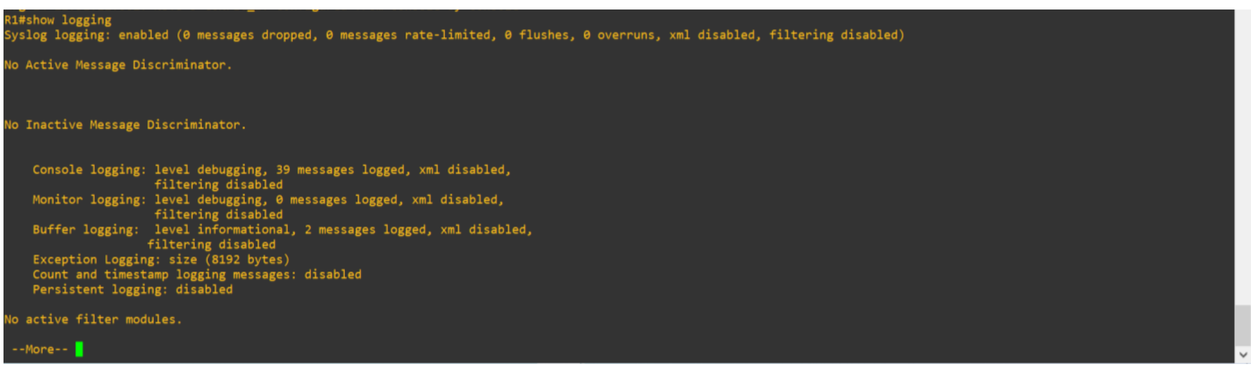
R1 was configured with buffered informational logging:

logging buffered 16384 informational
service timestamps log datetime msec

A harmless configuration change was also used to generate a log message for verification.

The show logging command was used to view the stored log messages.

Syslog


```
R1#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited, 0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.



No Inactive Message Discriminator.


    Console logging: level debugging, 39 messages logged, xml disabled,
                     filtering disabled
    Monitor logging: level debugging, 0 messages logged, xml disabled,
                     filtering disabled
    Buffer logging:  level informational, 2 messages logged, xml disabled,
                    filtering disabled
    Exception Logging: size (8192 bytes)
    Count and timestamp logging messages: disabled
    Persistent logging: disabled

No active filter modules.

 --More--
```

External Syslog Server Status

At this stage, there was **no external Syslog server attached to the GNS3 topology**.

Therefore:

Completed: Router-side/local logging

Pending: External Syslog server configuration

A dedicated server such as the GNS3 Networkers' Toolkit can be added later if the assignment specifically requires logs to be forwarded to an external Syslog server.

SNMPv3 Configuration

SNMPv3 was configured on R1 to provide secure network management using authentication and privacy.

The configuration included:

- SNMP view
- SNMPv3 group
- SNMPv3 user
- SHA authentication
- AES-128 privacy

The configuration used the following logical components:

View: SECUREVIEW

Group: SECUREGROUP

User: SNMPADMIN

The SNMPv3 user was successfully created and the router returned to configuration mode.

Basic Hardening

Basic router hardening was performed to reduce unnecessary exposure.

The following security controls were already implemented:

- SSH version 2
- Telnet disabled
- AAA
- Local user authentication
- Password encryption
- MOTD banner
- Secure VTY lines

R1 had only two active interfaces:

- FastEthernet0/0

- FastEthernet1/0

Both interfaces were required by the topology, so neither was disabled.

The service configuration was also reviewed.

The router contained:

service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption

These services were retained because they support logging and password security.

Connectivity Verification

Final connectivity verification was performed after the security configurations.

12.1 Ping Test

R1 successfully pinged R2 using:

ping 10.0.12.2

The successful replies confirmed connectivity across the WAN link.

Traceroute Test

Traceroute was performed from R1 to R2:

traceroute 10.0.12.2

The result showed:

1 10.0.12.2

R2 was reached directly as the first hop.

ACL / Security Verification

An extended ACL was configured on R1.

The ACL verification showed:

Extended IP access list 110

10 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255

The rule showed **11 matches**, which indicates that matching traffic had been processed by the ACL.

The verification command used was:

show access-lists

Zone-Based Firewall (ZBF)

Zone-Based Firewall was one of the requirements of Week 07.

The required ZBF components included:

- Security zones
- Zone pair
- Class map
- Policy map
- Service policy

The intended security design was:

LAN → R1 → WAN

with the LAN and WAN interfaces placed into separate security zones.

However, the available Cisco 7200 IOS image did not support the required Zone-Based Firewall commands.

The router was running:

Cisco 7200 Software

C7200-ADVIPSERVICESK9-M

Version 15.2(4)55

When the command:

zone security LAN

was entered, the router returned:

% Unrecognized command

The same issue occurred when checking R2.

Therefore, **ZBF has not been marked as completed** in this report.

This is an IOS feature limitation rather than a configuration error.

Requirement for Completion

To complete the ZBF portion, a Cisco router/IOS image supporting Zone-Based Firewall must be used.

The required commands would include concepts such as:

```
zone security LAN
zone security WAN
zone-pair security ...
class-map type inspect ...
policy-map type inspect ...
service-policy type inspect ...
```

The existing completed VPN, SSH, AAA, OSPF, and monitoring configurations should not be deleted when changing the lab environment.

Conclusion

Week 07 provided practical experience in implementing enterprise network security, secure management, VPN technologies, routing protection, monitoring, and network verification.

The IPsec VPN was successfully configured and verified, with the IKE security association reaching the **QM_IDLE / ACTIVE** state. SSH version 2, AAA, password encryption, secure VTY access, OSPF authentication, passive interfaces, and prefix-list filtering were also successfully implemented.

NTP was configured using R1 as the NTP master and R2 as the client. Router-side Syslog/logging was configured and verified, and SNMPv3 was configured with authentication and AES-128 privacy. Basic hardening was also reviewed, and final ping and traceroute tests confirmed WAN connectivity.

The remaining limitations are the **external Syslog server** and **Zone-Based Firewall**. The external Syslog server was not available in the original topology, while the current C7200 IOS image does not recognize the required ZBF commands. These limitations have been documented rather than falsely claiming completion.

Overall, the Week 07 implementation successfully covered the major enterprise security, management, monitoring, and verification requirements supported by the available GNS3 environment.

THE END